MeshCAD

User Manual and Practical Tutorials

2D block-structured mesh generator with integrated CAD



Manual version 1.0 | Application version 1.0

Contents

1. 1. Introduction
2. 2. Installation and startup
3. 3. Interface overview
4. 4. CAD drawing and editing
5. 5. Block meshing workflow
6. 6. Tutorial 1: rectangular block
7. 7. Tutorial 2: curved two-block domain
8. 8. Tutorial 3: biased distribution and mesh improvement
9. 9. Saving and VTK export
10. 10. Troubleshooting
11. 11. Quick reference

Recommended path Read Sections 3 to 5 first. Complete the tutorials in order because each tutorial reuses concepts introduced in the previous one.

1. Introduction

MeshCAD is a desktop application for creating 2D CAD geometry, organizing closed contours into structured blocks, assigning edge-node distributions, generating meshes through transfinite interpolation, and exporting multiblock results to VTK.

1.1 Main capabilities

- Draw line segments, circular arcs, and Catmull-Rom splines with live previews and vertex snapping.
- Edit topological vertices and internal control points by dragging or by entering coordinates.
- Detect closed loops and split a selected loop into four logical block sides.
- Assign uniform or biased node distributions to one or more edges.
- Enforce near-boundary orthogonality and apply elliptic smoothing to interior nodes.
- Save complete projects as JSON and export one VTS file per block through a PVD collection.

Core idea Geometry, topology, and mesh data are separate. Create a closed contour first, create a block from that contour, assign node counts, and only then generate the mesh.

1.2 Visual conventions

Element	Appearance	Meaning
CAD edge	Blue curve	Geometric boundary
Vertex	Light red circle	Shared topological endpoint
Grip	Light blue circle	Arc center or internal spline point
Selected loop	Thick green path	Active closed contour
Selected edges	Orange path	Selection in the Distributions panel
Mesh	Thin dark grid	Generated structured mesh

2. Installation and startup

2.1 Requirements

- Windows, Linux, or macOS.
- Python 3.9 or newer.
- PyQt5 as the runtime GUI dependency.
- pytest and pytest-qt for development and verification.

2.2 Install from the repository

Open a terminal in the project root and run:

```
python -m venv .venv
.venv\Scripts\activate
pip install -e .
```

Linux and macOS Activate the environment with: source .venv/bin/activate

2.3 Start MeshCAD

python -m meshcad

A packaged build can be started from dist/MeshCAD when the PyInstaller build is available.

3. Interface overview

3.1 Main areas

Area	Purpose
File menu	Create, open, save, save as, export VTK, and exit.
Toolbar	CAD tools, grid controls, fit view, and live mesh-improvement settings.
Central canvas	Drawing, selection, editing, zoom, pan, and mesh display.
Loops panel	Detect closed loops and create blocks.
Distributions panel	Select edges, set node counts, and choose a distribution.
Blocks panel	Display blocks, dimensions, and opposite-side mismatches.
Status bar	Operational messages and current cursor coordinates.

3.2 Toolbar controls

Control	Operation
Select	Leave the active drawing tool and enable geometry editing.
Line	Two clicks: start and end.
Arc	Three clicks: center, start, and end; traversal is counterclockwise.
Spline	Add points with left-click and finish with right-click.
Insert Point	Create an isolated vertex from X and Y coordinates.
Generate Mesh	Generate or update all valid block meshes.
Clear Mesh	Hide the mesh without removing CAD geometry or block definitions.
Show Grid / Spacing	Show the reference grid and set its spacing.
Reset Zoom	Fit all geometry and remove accumulated panning.
Orthogonal / Layers / Strength	Improve near-boundary alignment.

Control	Operation
Smoothing / Relaxation	Apply elliptic smoothing to interior nodes.

Tooltips Hold the pointer over any toolbar icon or control to display a concise explanation.

4. CAD drawing and editing

4.1 Canvas navigation

Action	Control
Zoom	Mouse wheel; zoom remains centered under the pointer.
Pan	Hold the middle mouse button and drag.
Fit complete model	Reset Zoom.
Cancel active tool	Esc.
Read coordinates	Status bar at the lower-right corner.

4.2 Create geometry

STEP 1 | Lines

Choose Line, click the first endpoint, and click the second endpoint. The dashed preview shows the candidate segment.

STEP 2 | Circular arcs

Choose Arc, click the center, click the start point to define the radius, and click the end point to define the terminal angle.

STEP 3 | Splines

Choose Spline, add control points with left-click, and finish with right-click. Internal points become editable grips.

Snapping When the cursor is close to an existing vertex, the new point snaps to it. Use snapping to guarantee topologically closed contours.

4.3 Edit geometry

- Drag a red vertex to move every connected edge.
- Drag a blue grip to move an arc center or an internal spline control point.
- Double-click a vertex or grip to enter exact X and Y coordinates.
- When an arc center is edited, the endpoints stay fixed and the center is projected onto the chord perpendicular bisector.
- If a mesh already exists, geometry changes trigger automatic regeneration.

5. Block meshing workflow



5.1 Detect a closed loop

Click Detect Loops. Select a listed loop to highlight it in green. If no loop appears, inspect endpoint connections and correct them through snapping or numerical editing.

5.2 Create a block

Select a loop, click Create Block from Loop, and select four distinct vertices. The selected vertices define the four logical corners. A block side may contain one or several CAD edges.

5.3 Node compatibility

Structured-grid condition Opposite sides 0/2 must have identical total node counts; sides 1/3 must also match. The Blocks panel reports OK or MISMATCH.

5.4 Edge distributions

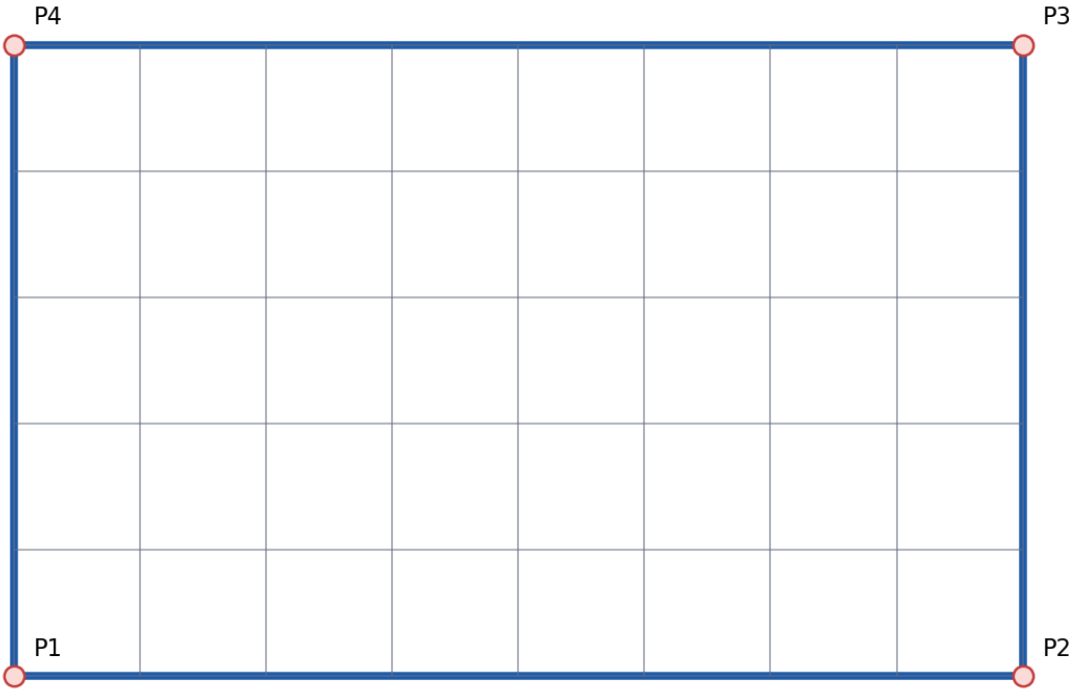
Type	Effect	Typical use
Uniform	Equal parameter spacing.	Regular domains and initial meshes.
Biased	Intervals follow a geometric progression.	Concentrate nodes near the edge start.
Bias = 1	Equivalent to uniform.	Neutral distribution.
Bias > 1	Small intervals first, larger intervals later.	Local resolution near a feature or wall.

5.5 Mesh improvement

Parameter	Meaning	Suggested start
Orthogonal	Enable correction normal to boundaries.	On for curved walls.
Layers	Number of corrected inner layers.	2 to 4.
Strength	Applied fraction of the correction.	0.70 to 0.90.

Parameter	Meaning	Suggested start
Smoothing	Number of elliptic iterations.	20 to 60.
Relaxation	Update fraction per iteration.	0.30 to 0.50.

6. Tutorial 1: rectangular block



Objective

Create a simple Cartesian mesh and verify node and cell counts.

STEP 1 | Create the contour

Draw consecutive line segments through (0,0), (4,0), (4,2.5), (0,2.5), and back to (0,0).

STEP 2 | Detect and select the loop

Click Detect Loops and select the four-edge loop.

STEP 3 | Create the block

Click Create Block from Loop and select the four corners.

STEP 4 | Set horizontal nodes

Select the bottom and top edges, set 9 nodes and Uniform, then apply.

STEP 5 | Set vertical nodes

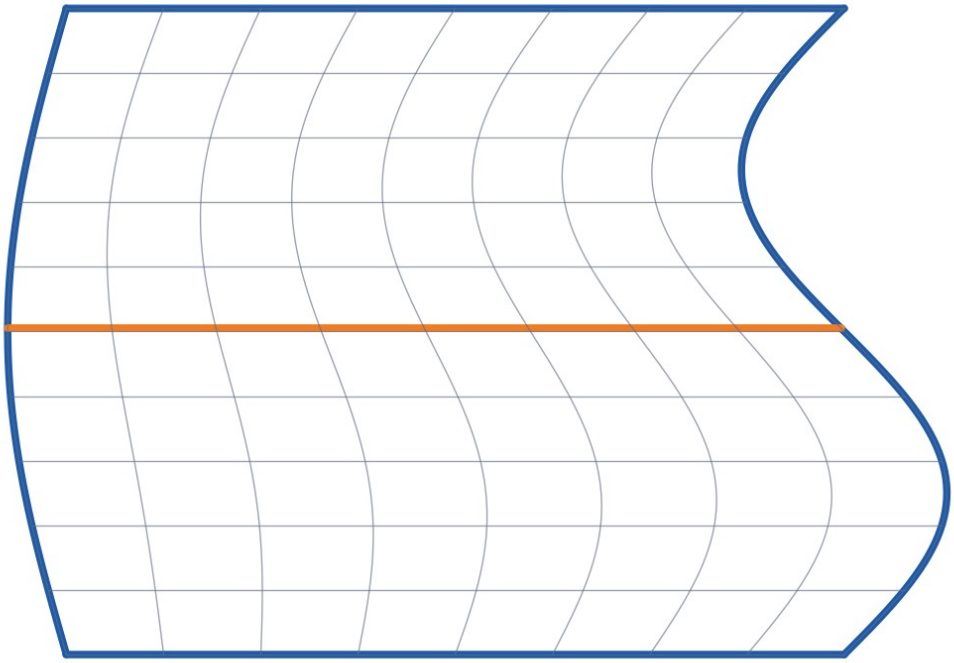
Select the left and right edges, set 6 nodes and Uniform, then apply.

STEP 6 | Generate

Click Generate Mesh. The result must contain 9 x 6 nodes and therefore 8 x 5 = 40 cells.

Verification The Blocks panel reports OK (nodes 9 x 6), interior lines are straight, and the status bar reports one block and 40 cells.

7. Tutorial 2: curved two-block domain



Objective

Build a curved domain, divide it into two conformal blocks, and verify a shared edge distribution.

STEP 1 | Outer contour

Draw the left and right boundaries with splines and close the top and bottom with lines.

STEP 2 | Internal division

Draw a transverse line between two existing boundary vertices. Snap both endpoints.

STEP 3 | Detect loops

Detect loops and identify the upper and lower subdomains.

STEP 4 | Create both blocks

Create one block from each subdomain and select four logical corners for each.

STEP 5 | Set distributions

Assign 15 uniform nodes to the principal edges. Confirm that both blocks report OK.

STEP 6 | Generate and edit

Generate the mesh, then drag a spline grip. Geometry and mesh update while the shared interface remains conformal.

Expected result Two blocks share exactly the same interface nodes. No gaps or duplicate interface lines should appear.

Using test.json

The supplied test.json project is a ready-made two-block example with seven edges and uniform node distributions. Open it, click Reset Zoom, and generate the mesh.

8. Tutorial 3: biased distribution and mesh improvement

Objective

Refine a region of interest and improve the orientation and smoothness of interior grid lines.

STEP 1 | Start from Tutorial 2

Open the curved domain or test.json and generate the uniform baseline mesh.

STEP 2 | Apply bias

Select an outer edge, choose Biased, set Bias to 1.5, and apply. Compare spacing at both ends.

STEP 3 | Increase resolution

Set 25 nodes in the direction requiring additional resolution. Confirm opposite-side compatibility.

STEP 4 | Enable orthogonality

Enable Orthogonal with 3 layers and Strength = 0.80.

STEP 5 | Smooth

Set Smoothing = 40 and Relaxation = 0.40. Live controls regenerate the mesh after a short debounce interval.

STEP 6 | Compare

Temporarily disable Orthogonal and set Smoothing to 0. Restore the combination that produces the cleanest grid.

Visual quality criteria

- No crossed or visibly inverted cells.
- Gradual transitions between small and large cells.
- Smooth interior grid lines without oscillations.
- Near-right angles along boundaries where orthogonality matters.
- Exact continuity across adjacent blocks.

Caution A large bias combined with too few cells can create abrupt size transitions. Increase Bias gradually and inspect every change.

9. Saving and VTK export

9.1 JSON projects

File > Save stores geometry, topology, edge distributions, blocks, and mesh-improvement settings. Save As creates a separate project copy. JSON is readable and version-control friendly, but editing through MeshCAD is recommended.

9.2 Export VTK

STEP 1 | Generate the mesh

Export is available only after a mesh has been generated.

STEP 2 | Export

Choose File > Export Mesh (VTK) and select a .pvd filename.

STEP 3 | Keep the output together

MeshCAD writes the PVD and a <name>_blocks folder containing one VTS file per block.

STEP 4 | Inspect

Open the PVD file in ParaView to load every block together.

Important Do not separate the PVD file from its _blocks folder because the PVD uses relative paths.

10. Troubleshooting

Symptom	Likely cause	Solution
No loops detected	Open contour or non-coincident endpoints.	Zoom in and edit the affected vertices numerically.
Block reports MISMATCH	Opposite sides have different totals.	Apply compatible node counts.
An edge cannot be deleted	The edge belongs to a block.	Delete the associated block first.
No mesh appears	No valid block or incompatible counts.	Review the Blocks panel and generate again.
Geometry is outside the view	Accumulated zoom or panning.	Click Reset Zoom.
Arc center differs from input	Center must remain equidistant from endpoints.	The center is projected onto the perpendicular bisector.
Mesh is strongly distorted	Excessive bias, strength, or relaxation.	Reduce parameters and increase them gradually.
VTK export unavailable	No generated mesh exists.	Generate the mesh first.

11. Quick reference

11.1 Mouse and keyboard

Input	Action
Left-click	Place points, select items, or begin dragging.
Double left-click	Edit vertex or grip coordinates.
Right-click	Finish a spline.
Middle drag	Pan the view.
Mouse wheel	Zoom under the pointer.
Esc	Cancel the active tool.

11.2 Minimal workflow

1. Draw a closed contour.
2. Detect and select its loop.
3. Create a block by selecting four corners.
4. Assign node counts and distributions.
5. Confirm that the block reports OK.
6. Generate the mesh.
7. Save the project and export VTK if required.

11.3 Recommended starting values

Case	Nodes	Orthogonal	Smoothing	Relaxation
Rectangle test	9 x 6	Off	0	-
Simple curved block	15 x 15	2 layers, 0.80	30	0.40
Interactive adjustment	Low to medium	Optional	10-20	0.30
Final mesh	As required	As required	40-100	0.30-0.50

Final checklist Closed contours · blocks report OK · conformal interfaces · no crossed cells · project saved · export verified.